Taehoon Hwang

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Education

Columbia University — New York, NY
B.S. in Computer Science; GPA: 4.05 / 4.33 — *Aug. 2025 – Present*
Relevant Courses *Computer Science Theory, Programming Languages and Translators, Introduction to Databases, Projects in Computer Science, Theory ML Interaction, Natural Language Processing*

Purdue University — West Lafayette, IN
B.S. in Computer Science, Minor in Mathematics; GPA: 4.00 / 4.00 — *Aug. 2023 – May 2025*
Relevant Courses *Linear Algebra, Probability, Discrete Mathematics, Computer Architecture, Data Structures, Statistical Methods, Competitive Programming I & II, Foundations of Deep Learning, Data Mining & Machine Learning, Introduction to Artificial Intelligence*

Experience

Artificial Intelligence Researcher — Jun. 2026 – Present
Asteromorph — *Seoul, South Korea*
- Working on deploying scalable and efficient reinforcement learning engines for use in agentic systems

Research Assistant — Sep. 2025 – Present
Columbia University — *New York, NY*
- Working on fine-grained advantage computations for reinforcement learning to improve GRPO methodologies

Artificial Intelligence Research Intern — May 2025 – Aug. 2025
Asteromorph — *Seoul, South Korea*
- Worked with fine-tuning and hosting LLMs (>200B parameters) on multi-GPU servers using vLLM and llama.cpp
- Developed an LLM fine-tuning framework by writing kernel code with PyTorch, Unsloth, and Hugging Face
- Deployed a parallel LLM fine-tuning scheduler and metric dashboard using Ray, Grafana, Prometheus, and Docker

Teaching Assistant — Jan. 2025 – May 2025
Purdue University — *West Lafayette, IN*
- Held office hours and lab sessions for students in Discrete Mathematics and Programming in C courses
- Reviewed and conducted problem solving sessions for discrete mathematics on first order logic, set theory, etc
- Answered students' questions during lab sessions on UNIX systems, file IO, dynamic memory allocation, etc

Research Assistant — Jan. 2024 – Dec. 2024
Purdue University — *West Lafayette, IN*
- Engineered demand forecasting pipeline integrating probability curves with LSTM encoding for enhanced accuracy
- Optimized inference storage by achieving a reduction of >99% compared to traditional deep learning models
- Conducted latent space analysis and illustrated composite encoding of demand features using LSTM models

Machine Learning Intern — Sep. 2023 – Aug. 2024
Quantum Research Sciences — *West Lafayette, IN*
- Employed ML algorithms and statistical techniques to deploy software for the United States Air Force
- Produced Monte Carlo simulations to test and tune quantum algorithms for inventory management
- Developed end-user interface for data analytics in quantum inventory management software

Research Assistant — May 2021 – Dec. 2021
Sungkyunkwan University — *Seoul, South Korea*
- Engaged with research team to boost open set recognition in ResNet models by 9%p via latent space manipulation
- Developed and deployed a remote automated testing and optimization framework for models using PyTorch
- Presented research findings at the Korean Science High School R&E Conference

Publications & Contributions

LiveNewsBench: Evaluating LLM Web Search Capabilities May. 2026
Yunfan Zhang, Kathleen McKeown, Smaranda Muresan (Contributor) *ICML 2026*

Spacer: Towards Engineered Scientific Inspiration Aug. 2025
Minhyeong Lee et al. (Co-author) *arXiv Preprint*

Projects

Graduate School Research Agent Apr. 2026 – Present

- Developed agentic system to research graduate school programs, including fitness, quality, and difficulty
- Implemented agentic retrieval loop, LLM quality judging, CV-aware fit assessment, and gap-fill passes using the Anthropic SDK
- Added rich reporting using markdown files, full trajectory viewing and analysis, and interactive UIs

Stack Rotation May 2025 – Aug. 2025

- Developed full-rank fine-tuning LoRA methodology with PyTorch using stochastically sampled rotation matrices
- Achieved SOTA-comparable performance when compared to LoRA variants such as DoRA, VeRA, and HiRA
- Constructed parallel fine-tuning orchastrator using Ray, Grafana, Prometheus, and Docker

Rotationally Equivariant Spatio-temporal Prediction Feb. 2025 – Apr. 2025

- Developed spatio-temporal predictive models with rotational equivariance to adapt to out-of-distribution inputs
- Constructed training infrastructure using PyTorch Lightning and distributed high-performance computing nodes
- Compared methods for equivariance using steerable wavelet filters, G-CNNs, and equivariant attention modules

Financial Anomaly Detection with Modified Benford's Law Jan. 2025 – Mar. 2025

- Suggested novel alteration of Benford's Law applicable to stock return data based on the Student-Lévy process
- Proved algorithm utility with real-life stock data and demonstrated 60% improvement over conventional methods
- Utilized new algorithm to detect financial anomalies in stock returns based on fitted location-scale t-distributions

WUMT: Wavelet U-Net Motion Transformer Jul. 2024 – Mar. 2025

- Investigated spatio-temporal encoding methods of videos using discrete wavelet transforms and NAFNet blocks
- Utilized 4D motion tensor computations and predictions with transformers for U-Net latent space operations
- Built training infrastructure with PyTorch Lightning and leveraged Docker and MLFlow for streamlined research

Contextual-Diffusion Feb. 2023 – Aug. 2023

- Improved cohesion and features of Stable Diffusion model output utilizing spacial context from LLMs
- Developed pipeline incorporating Mask-RCNN models as translation layers between LLMs and Stable Diffusion
- Generated and annotated image segmentation datasets for supervised learning of Mask-RCNN models

Minimax-based Animal Shogi AI Mar. 2021 – Nov. 2021

- Programmed an Animal Shogi bot using a minimax algorithm with alpha-beta pruning in C++
- Implemented interactable game GUI and cross-language translation mechanism with Python
- Demonstrated bot with live play-testing at school festival to 350+ students and faculty

Activities

Purdue Hackers Club Member Jan. 2025 – May 2025
Purdue University *West Lafayette, IN*

- Participated in weekly 'hack' sessions and developed projects involving full-stack web development.
- Developed website for course scheduling at Purdue with Next.js, MongoDB, and Python among team of 3.

Purdue IGDC Club Member Jan. 2024 – May 2025
Purdue University *West Lafayette, IN*

- Participated in various in-person game development sessions with other club members.
- Developed game with Unity centered around creating gravitational wells to enduce fast-paced gameplay.
- Gave feedback on other games being developed and participated in play testing.

ICPC 2023 ECNA Regional Participant Sep. 2023
Purdue University *West Lafayette, IN*

- Ranked in the Top 20 in Purdue's ICPC participant selection contest
- Lead team of 3 in the ICPC East Central North America regional contest among 5 other Purdue ICPC teams
- Solved problems utilizing competitive programming techniques such as dynamic programming and backtracking

Hello World Hackathon Participant Sep. 2023
Purdue University *West Lafayette, IN*

- Led a team of 4 in a 24-hour hackathon driving project development and collaboration
- Built a software pipeline utilizing LLMs to deliver personalized dietary text feedback to users
- Built a full-stack web app with a React front-end and ExpressJS back-end integrating MongoDB RestAPIs

ML@Purdue Club Member Mar. 2023 – May 2025
Purdue University *West Lafayette, IN*

- Engaged in club discourse on AI and ML topics for research and software development.
- Participated in paper reading and workshops on various topics, such as transformers, GNNs, etc.

Published Author on Neural Network Fundamentals Sep. 2021 – Mar. 2022
Barun Books Co.,Ltd *Seoul, South Korea*

- Authored a 160-page book on neural networks covering backpropagation, gradient descent, and parallelization
- Distributed to 13 retailers, selling 300+ copies in the first year
- Ranked as a top 4 entry in the "Weekly Top Releases" by the second-largest South Korean book retailer

Honors & Awards

Dean's List
Columbia University, 2025 & 2026

Purdue Office of Undergraduate Research Grant
Purdue University, 2024 *$ 500*

Dean's List & Semester Honors
Purdue University, 2023 & 2024 & 2025

Excellence Award in the Gifted School Research Conference
Pohang University of Science & Technology, 2022

Azure AI Fundamentals Certification
Microsoft, 2022

Technical Skills

Languages & Frameworks: C/C++, Express.js, Flask, HTML/CSS, Java, JavaScript, Python, React, x86 Assembly
Dev & Research Tools: Docker, Git/GitHub, IntelliJ, LaTeX, MLfLow, Overleaf, PyCharm, Slurm, VS Code
Libraries: Keras, NumPy, OpenCV, Pandas, PyTorch, Scikit-learn, TensorFlow, Transformers, Unsloth, trl, vLLM